

On really big numbers

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- ▶ Modern times
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History of writing numbers

How big numbers people needed in:

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- ▶ empires
- ▶ modern times - let's discuss that later

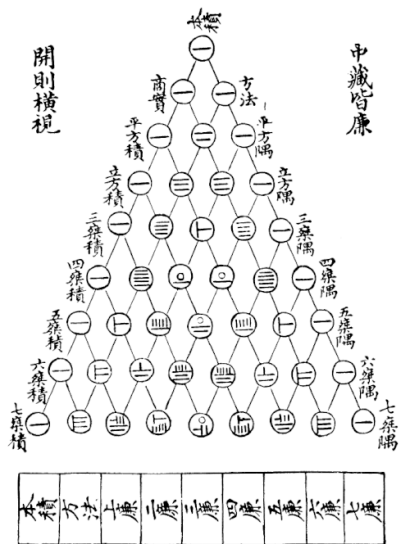
History of writing numbers - Babylon

 1	 11	 21	 31	 41	 51
 2	 12	 22	 32	 42	 52
 3	 13	 23	 33	 43	 53
 4	 14	 24	 34	 44	 54
 5	 15	 25	 35	 45	 55
 6	 16	 26	 36	 46	 56
 7	 17	 27	 37	 47	 57
 8	 18	 28	 38	 48	 58
 9	 19	 29	 39	 49	 59
 10	 20	 30	 40	 50	

History of writing numbers - Maya

0	1	2	3	4
	•	••	•••	••••
5	6	7	8	9
	• 	•• 	••• 	•••• 
10	11	12	13	14
	• 	•• 	••• 	•••• 
15	16	17	18	19
	• 	•• 	••• 	•••• 

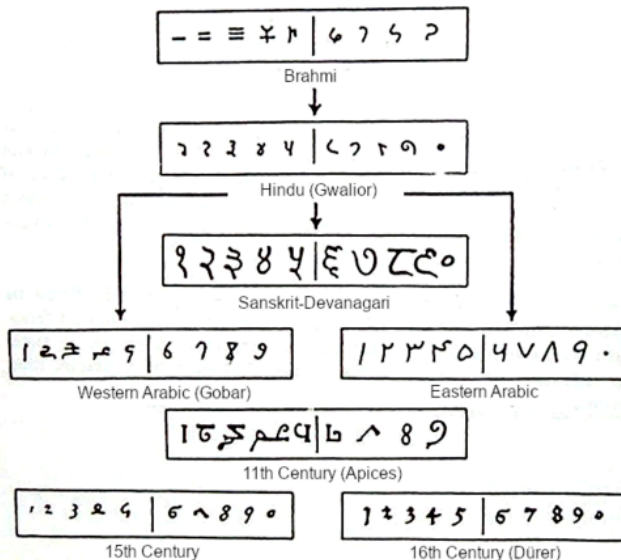
古法七藥方圖



History of writing numbers - Roman Empire



History of writing numbers - Indian → Arabic



History of writing numbers - Modern times

Positional decimal system

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- ▶ $2 \times 10^{-9} = 0.000000002$ – small number

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trillions – $1,000,000,000,000 = 10^{12}$

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$7\mu\text{m} = 7 \times 10^{-6}\text{m}$ - the size of a cell nucleus

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$< 10^{-19} \text{ m}$ – the size of an electron

Conclusion: The positional decimal system is sufficient!

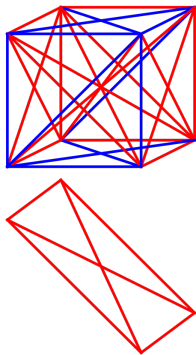
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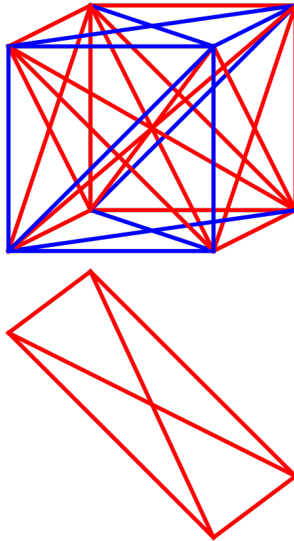
Problem: Connect each pair of geometric vertices of an n -dimensional hypercube to obtain a complete graph on 2^n vertices. Colour each of the edges of this graph either red or blue. What is the smallest number n for which every such colouring contains at least one single-coloured complete subgraph on four coplanar vertices?

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where G is *Graham's number*. So what is Graham's number?

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- ▶ $2 \uparrow^n 4 = 2 \uparrow^{n-1} (2 \uparrow^{n-1} (2 \uparrow^{n-1} 2))$ – madness!

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GRAHAM'S NUMBER

$$G = G_{64} = 3 \uparrow^{G_{63}} 3.$$

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Any questions?

Further reading



Thank you!

Primary Lunch

Secondary Lunch